

Prevalence and trends of double burden of malnutrition at household-level among mother-child pairs in South and Southeast Asia

Authors: Ashis Talukder^{1,2}, Matthew Kelly¹, Darren Gray^{1,3}, Haribondhu Sarma¹

Affiliations:

¹National Centre for Epidemiology and Population Health, Australian National University, Canberra, ACT 2600, Australia

²Statistics Discipline, Khulna University, Khulna-9208, Bangladesh

³Director of Population Health at the QIMR Berghofer Medical Research Institute, Australia

Corresponding Authors' email: Ashis.Talukder@anu.edu.au

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Summary

Background: The double burden of malnutrition (DBM), which involves both undernutrition and overweight/obesity, has become a significant global health concern. Despite the growing research on DBM, existing studies often focus on single countries or specific populations, leaving significant gaps in understanding its prevalence and trends among mother-child pairs in South and South-east Asia. Therefore, this study aims to estimate the prevalence and trends of DBM using the most up-to-date nationally representative data from multiple countries in these regions.

Methods: We analysed cross-sectional data from the Demographic and Health Survey, conducted between 2004 and 2022, for Bangladesh, Cambodia, Nepal, and Timor-Leste, including households with at least one mother-child pair. Using survey weights, we calculated the weighted prevalence of DBM, underweight children, and overweight mothers for each country. We examined associations between DBM prevalence and factors like wealth index, residence, and child sex by using chi-square tests. We assessed socioeconomic disparities using odds ratios obtained from a multivariable logistic regression model.

Findings: Prevalence of DBM in Bangladesh increased from 4% in 2004 to 13.8% in 2018, with underweight children declining from 47.6% to 21.5% and overweight/obese mothers rising from 11.6% to 42.3%. We observed similar trends in Cambodia (DBM prevalence: 7.4% in 2005 to 14.3% in 2022; underweight child: 28.3% in 2005 to 16.5% in 2022; overweight/obese mother: 18.6% in 2005 to 48.8% in 2022), Nepal (DBM prevalence: 5.6% in 2006 to 12.3% in 2022; underweight child: 38.6% in 2006 to 19% in 2022; overweight/obese mother: 14.4% in 2006 to 48.7% in 2022), and Timor-Leste (DBM prevalence: 9.4% in 2009 to 15.8% in 2016; underweight child: 44.1% in 2009 to 40.8% in 2016; overweight/obese mother: 15.1% in 2009 to 25.6%). In all four countries, households with higher socioeconomic status consistently show a greater likelihood of experiencing DBM (odds ratio > 1) compared to lower-status households. Additionally, mother-child pairs residing in rural areas demonstrate a lower likelihood of experiencing DBM compared to those in urban areas.

Interpretation: DBM is on the rise in households across South and Southeast Asian nations, particularly in urban areas and among wealthier households. Despite improvements in reducing underweight prevalence, persistent undernutrition remains a significant challenge in the region. Simultaneously, the rising prevalence of obesity is a growing concern. Future research should prioritize identifying country-specific risk factors for DBM and understanding factors contributing to high rates of maternal overweight.

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Keywords Double burden of malnutrition, South and Southeast Asia, Prevalence, Trends

Research in context

Evidence before this study

We searched PubMed, Scopus and Google Scholar to identify relevant studies of double burden of malnutrition using the keywords: “double burden of malnutrition”, “dual burden of malnutrition”, “nutrition transition”, “prevalence of double burden of malnutrition”, “double burden households”, “overweight mother”, “undernourished child”, “South Asia”, “Southeast Asia”, “Bangladesh”, “Cambodia”, “Nepal”, “Timor-Leste”. While research on DBM has increased, significant gaps persist in estimating its prevalence and trends among South and Southeast Asia. Most studies in these areas only examine the nutritional status of either the mother or the child, with few taking a comprehensive regional perspective.

Added Value of This Study

This study provides a comprehensive cross-country analysis of household-level DBM prevalence and trends in Bangladesh, Cambodia, Nepal, and Timor-Leste, revealing consistent increases in DBM across all four countries. Our findings highlight a rising prevalence of overweight or obese mothers alongside a declining prevalence of underweight children, suggesting that maternal overweight is a significant factor in DBM trends. This study also identified that there exist socioeconomic and urban-rural disparities of DBM prevalence among these countries.

Implications of All the Available Evidence

Increased consumption of ultra-processed foods and decreased physical activity lead to an increase in overweight mothers, while inadequate infant feeding and a lack of healthy foods at home may be the causes of persistent child undernutrition in households. The continuation of undernourished children and the increasing number of overweight mothers in the same households further increase the prevalence of DBM at the household level in the studied countries. The findings of this study indicate that wealthier households experience a higher DBM burden, possibly due to high-calorie diets and sedentary lifestyles, although the reasons for child undernutrition in rich families are unclear. Urban households experience greater DBM challenges than rural ones, possibly due to less physically demanding lifestyles and greater access to processed foods. Further research is critical for developing effective strategies for addressing DBM and exploring its complex socioeconomic determinants.

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Introduction

The double burden of malnutrition (DBM) is the simultaneous presence of undernutrition (underweight, wasting, stunting, and micronutrient deficiencies) along with overnutrition (obesity or overweight) at various population levels, such as country, community, individual, and household [1]. DBM is frequently observed in households where a mother is overweight or obese and her child suffers from undernutrition, such as stunting, wasting, or being underweight [1]. The primary cause of this dual burden is the nutrition transition [2], which is associated with major shifts in the diet towards more refined carbohydrates, added sugars, edible oils, and foods derived from animals, as well as a decrease in physical activity, contributing to overweight or obesity in adults [3]. This major change in dietary behaviour eventually led to a shift from being mostly undernourished to having a higher prevalence of overweight or obesity [4]. Moreover, a combination of inadequate infant feeding practices, poor sanitation and hygiene, and limited access to nutritious foods may lead to an increased prevalence of undernutrition among children [5,6].

Global prevalence of DBM is increasing. A recent study using data from 126 countries between 1999 and 2017 highlighted that the prevalence of total household-level DBM ranges between 3% and 35% [2]. During this period, the most common DBM pairs were stunted child-overweight mother pairs (1–24%), followed by overweight mother-wasted child pairs (0.3–10%), and underweight mother-overweight child pairs (0.1–4%) [2]. DBM is particularly concerning in many South and Southeast Asian countries. In 2020, a systematic review and meta-analysis found that 28% of the South Asian population was underweight and 17% was overweight, while in Southeast Asia, the rates were 20% for both conditions [15]. Another study across eight South and Southeast Asian countries found a pooled prevalence of 10.0% for overweight mother-stunted child pairs, 7.0% for overweight mother-wasted child pairs, and 7.0% for overweight mother-underweight child pairs [7]. Additionally, this study revealed a 12.0% prevalence for any form of DBM coexistence (overweight mothers and any form of child undernutrition) at the household level [7]. Over time, India has seen an increase in household-level DBM, affecting 1 in 4 households in 2021 [16]. In Nepal, one study reported a 6.6% prevalence of DBM among mother-child pairs [17]. In Cambodia, one in three children under five years old suffers from stunting, with approximately one-fourth of these stunted children having an overweight or obese mother [18].

Household-level DBM has significant health and economic consequences. In terms of health, households with DBM experience immediate health impacts, such as stunted growth in children due to undernutrition, along with long-term obesity-related conditions in adults, including type 2 diabetes, hypertension, and certain cancers [4, 8]. This dual burden also increases the risk of chronic diseases, as children who experience stunting may become overweight in adulthood [11–13]. Managing these diverse health issues often requires costly medical interventions, which impose a substantial financial burden on families and increase national healthcare costs [4]. The economic impact is further worsened by reduced productivity due to illness, leading to lower household income [4]. Additionally, the cost of managing both undernutrition and obesity within the same household can

lead to increased out-of-pocket expenses [14]. This financial strain can push families into deeper poverty, which continues the cycle of poverty and malnutrition. On a larger scale, DBM also hinders a country's economic development by reducing its gross national product [4].

While research on DBM has gained attention globally, significant gaps remain in estimating its prevalence and trends among mother-child pairs in South and Southeast Asia. Existing studies on DBM often focus on single-country analyses or specific population groups [19–22], limiting our understanding of the broader regional context. Additionally, most studies in these regions use small-scale samples [45–48] and focus separately on the nutritional status of either the mother or the child, rather than both simultaneously. While some studies have estimated DBM prevalence at the household level within individual countries in South and Southeast Asia, few have taken a region-wide perspective and often rely on outdated datasets [7, 23]. Therefore, this study aims to estimate the prevalence and trends of DBM using the most up-to-date nationally representative data from multiple countries in South and Southeast Asia.

Methods

Study design and Study area

We conducted a multi-country secondary data analysis focusing on four countries in South and Southeast Asia: Bangladesh, Cambodia, Nepal, and Timor-Leste. These countries were selected because they are experiencing a nutritional transition from healthy to unhealthy processed foods, leading to the dual challenge of undernutrition and overnutrition, which worsens the issue of DBM [24]. Additionally, these countries exhibit varying levels of maternal and child undernutrition, providing a diverse context for analysis [25]. Most importantly, all the necessary data on the nutritional status of both mothers and children was available in these selected countries.

Data sources

We extracted necessary information from nationally representative cross-sectional data from the Demographic and Health Surveys (DHS), conducted between 2004 and 2022, for four South and Southeast Asian countries, Bangladesh (2004, 2007, 2011, 2014, 2018), Cambodia (2005, 2010, 2014, 2022), Nepal (2006, 2011, 2016, 2022), and Timor-Leste (2009–10 & 2016). DHS used a two-stage, stratified random sampling method. First, census enumeration areas (EAs) were chosen based on a probability proportional to size approach. Households were then randomly selected from the EA. Data were collected by face-to-face interviews with women aged 15 to 49 years and their children aged 0 to 5 years within these selected households. Participants provided voluntary written informed consent before interviews. Ethical approval was granted by both the ICF International institutional review board in Calverton, MD, USA, and individual review boards in each participating country. The DHS dataset is accessible from the DHS programme website, in compliance with a data-sharing policy. Comprehensive descriptions of DHS sampling procedures, questionnaire validation, and data collection methods are available on the DHS website [26].

Study population

In this study, we included households, each containing at least one mother-child pair. We then considered mother-child pairs from households that had information on both the mother's and her child's nutritional status. This resulted in different samples for each country and year. Samples missing information on child's nutrition status or mother's BMI were excluded. Details on the final sample included for each country and each time point are in Figure 1.

Variables

The DHS used anthropometric measurements (weight and height/length) to investigate the nutrition status of under-five children. Key indicators such as height-for-age, weight-for-age, and weight-for-height were used to evaluate the nutritional health of the participants. To determine the nutritional status of children, we applied the 2006 World Health Organisation (WHO) growth standard to calculate anthropometric z-scores. Based on these z-scores, children were classified into three categories: stunted (height-for-age z-score < -2 standard deviation), underweight (weight-for-age z-score < -2 standard deviation), and wasted (weight-for-height z-score < -2 standard deviation) [27,28]. Maternal body mass index (BMI) was calculated as weight (kg)/height (m²). We classified mothers into underweight (BMI < 18.5 kg/m²), normal (BMI: 18.5-22.9 kg/m²), and overweight/obese (BMI ≥ 23 kg/m²) categories using Asian-specific BMI cutoffs [7, 29]. To determine household-level DBM, we identified households where the mother was overweight or obese and the child was undernourished, categorised as underweight, stunted, or wasted based on the three indicators mentioned above. Subsequently, we created a composite outcome variable indicating households where an overweight or obese mother coexisted with at least one form of child undernutrition, including being underweight, stunting, or wasting.

DBM: In the same household, there is an overweight or obese mother and a stunted, wasted, or underweight child.

We consider three exposure variables, including place of residence (urban or rural), child's sex (male or female), and household wealth index. The DHS programme used principal component analysis (PCA) to categorise the wealth index, which represents a household's overall living standard, into five quintiles: poorest, poorer, middle, richer, and richest. Initially, the DHS programme computed a wealth score based on household characteristics via PCA and subsequently grouped households into quintiles based on this score [26].

Statistical analysis

We calculated the weighted prevalence of DBM, underweight children, and overweight mothers for each country and available time points, considering the clustered sampling design and survey weights. Then, we conducted a descriptive analysis to outline the characteristics of the study population. Using the chi-square test, we examined the significant association between DBM prevalence and various factors like wealth index, place of residence, and child sex. To assess socioeconomic disparities, we used odds ratios from a multivariable logistic regression

model that used maximum likelihood estimation to find out how each predictor variable affected the chance of a mother-child pair developing DBM. We calculated odds ratios for extreme categories in the household wealth index (lowest quintile vs highest quintile).

Results

The final analytical sample size for each country for each year and the frequency distribution for key characteristics among mother-child pairs from Bangladesh, Cambodia, Nepal, and Timor-Leste are shown in Table 1. The percentage of urban pairs in Bangladesh varied from 30.1% in 2004 to 34.9% in 2018. In Cambodia, the proportion increased from 20.3% in 2005 to 35.2% in 2022. The percentage of urban pairs varied in Nepal, from 22.5% in 2006 to 49.5% in 2022. Similarly, in Timor-Leste, the proportion varied from 21.6% in 2009 to 29.2% in 2016. In 2022, Nepal had the largest percentage of mother-child pairs living in the poorest families, with a rate of 33.2%. In contrast, Bangladesh had the lowest proportion, with around 20% in 2007. On the other hand, the percentage of mother-child pairs residing in the wealthiest families was lowest in Nepal (11.3% in 2022) and highest in Cambodia (24% in 2014). The sex distribution of children is largely equitable across countries and over time.

Figure 2 illustrates the trends in DBM prevalence, underweight children, and overweight/obese mothers across different survey years in four countries. In Bangladesh, DBM prevalence has consistently increased from 4% in 2004 to 13.8% in 2018. During the same period, the prevalence of underweight children in Bangladesh decreased from 47.6% to 21.5%, while the prevalence of overweight or obese mothers increased from 11.6% to 42.3%. We observed similar trends in Cambodia (DBM prevalence: 7.4% in 2005 to 14.3% in 2022; underweight child: 28.3% in 2005 to 16.5% in 2022; overweight/obese mother: 18.6% in 2005 to 48.8% in 2022), Nepal (DBM prevalence: 5.6% in 2006 to 12.3% in 2022; underweight child: 38.6% in 2006 to 19% in 2022; overweight/obese mother: 14.4% in 2006 to 48.7% in 2022), and Timor-Leste (DBM prevalence: 9.4% in 2009 to 15.8% in 2016; underweight child: 44.1% in 2009 to 40.8% in 2016; overweight/obese mother: 15.1% in 2009 to 25.6%). Although DBM is increasing over time for all countries, the overweight mother-stunted child has a higher prevalence than the other two types of DBM combination (overweight mother-wasted child and overweight mother-underweight child), indicating that stunting is more important to drive DBM prevalence than being underweight or wasted (Supplementary Figure 1).

Table 2 illustrates the prevalence of DBM across selected background variables in all the countries studied, alongside significance levels obtained from the chi-square test. In all countries, the most recent survey showed a high prevalence of DBM across various categories of each covariate. In Bangladesh, for instance, the prevalence of DBM was highest in urban areas (15.8%) and wealthier households (16.4%) in 2018. We observed similar patterns in Nepal, with higher rates of DBM in wealthy households (12.5%) and urban areas (14.2%) in 2022. Moreover, the prevalence of DBM in Cambodia was also higher in urban areas (19.4%) and wealthier families (16.5%) in 2022. In Timor-Leste, urban areas (20%) and richest households (20%) had greater DBM

prevalence in 2016. Notably, in Bangladesh and Timor-Leste, household-level DBM was significantly associated with place of residence and wealth index across all years ($p < 0.05$). In Cambodia, these factors were significant in 2005, 2010 and 2022. Conversely, in Nepal, wealth index was significantly associated with DBM across all years, while place of residence was significant only in 2006 and 2022 ($p < 0.05$).

Figure 3 shows the trends in disparities in DBM prevalence among mother-child pairs in four countries, based on wealth index, place of residence, and child sex. In all four countries, households with a higher wealth index consistently have a greater likelihood of experiencing DBM (odds ratio > 1) compared to those with a lower wealth index. For instance, in Bangladesh in 2004, households in the highest wealth quintile were 2.71 times more likely (95% CI: 2.01, 3.66) to face DBM compared to those in the lowest quintile. By 2018, this likelihood decreased but remained significant, with households in the highest quintile being 1.19 times more likely (95% CI: 1.03, 1.36) to experience DBM. From the earliest to the most recent survey years, we observed similar trends in Cambodia, Nepal, and Timor-Leste, where higher wealth quintiles were associated with a greater likelihood of DBM. This indicates the presence of socioeconomic disparities in all these countries. Figure 3 also highlights that mother-child pairs in rural households have a lower likelihood of experiencing DBM compared to those in urban areas [Bangladesh (2004): OR=0.55 (CI: 0.41, 0.72); Bangladesh (2018): OR=0.67 (CI: 0.58, 0.77); Cambodia (2005): OR=0.71 (CI: 0.53, 0.94); Cambodia (2022): OR=0.46 (CI: 0.37, 0.56); Nepal (2006): OR=0.79 (CI: 0.60, 1.04); Nepal (2022): OR=0.72 (CI: 0.54, 0.95); Timor-leste (2009): OR=0.79 (CI: 0.71, 0.98); Timor-leste (2016): OR=0.63 (CI: 0.53, 0.76)]. This relation indicates that the disparities exist between rural and urban regions in each country. However, we observed no notable patterns in relation to child sex across the nations.

Discussion

We performed a multi-country analysis to estimate the prevalence and trends of household-level DBM by using data from Bangladesh, Cambodia, Nepal, and Timor-Leste. Results indicate that the prevalence of DBM increased in all countries: from 4% to 13.8% in Bangladesh (2004–2018), 7.4% to 14.3% in Cambodia (2005–2022), 5.6% to 12.3% in Nepal (2006–2022), and 9.4% to 15.8% in Timor-Leste (2009–2016). Although the prevalence of underweight children decreased, the proportion of overweight or obese mothers increased substantially. DBM was notably higher in urban areas and rich households across all countries. We found significant associations between DBM, place of residence, and wealth index. Rich households continuously had higher likelihood of DBM indicating that urban-rural disparities were prominent, with urban areas showing higher DBM prevalence.

In all four countries, we observed similar trends: increasing DBM prevalence, declining the prevalence of underweight children, and rising overweight or obese mothers, indicating that the prevalence of overweight mothers may contribute to the rising prevalence DBM which aligns with prior studies [7, 30-32]. Several societies are shifting towards diets which is high in processed foods, unhealthy fats, sugars and refined

carbohydrates that contributing to the rising rates of overweight and obesity as well as decreased physical activity [2, 3]. Despite the decreasing prevalence of underweight children, the persistence of undernutrition remains a critical issue, exacerbated by factors such as limited access to nutritious foods, inadequate feeding practices, and susceptibility to infectious diseases [33]. Furthermore, in many societies, there exists a societal norm that compels women to give priority to their own dietary requirements, which may result in excessive eating and contribute to the growing prevalence of overweight and obesity [34]. Similarly, cultural restrictions that restrict children's consumption of certain nutrient foods can increase malnutrition issues. However, more research is necessary to know the actual reasons for these trends to the increasing prevalence of DBM [30].

Our study revealed that mother-child pairs residing in wealthy households have a higher risk of DBM in comparison to those residing in poor households. One possible reason for these disparities might be the dietary practices of the mothers in wealthy households. Mothers from wealthy households generally choose meals that are insufficient in essential nutrients and rich in fat, therefore promoting sedentary behaviours and increasing their potential to develop overweight or obesity [36]. Furthermore, studies suggest that mothers from wealthier backgrounds are more likely to be overweight or obese and less likely to be underweight [37–40]. While existing literature extensively suggests that children from poorer households are more likely to have underweight conditions compared to their wealthier counterparts [37–42], understanding why undernutrition persists among children in rich households requires further investigation.

We found significant differences between urban and rural regions in each of the four countries. Compared to households in urban areas, those in rural areas usually have a lower DBM burden. A variety of factors could potentially cause this phenomenon. Primarily, we anticipate that the physically demanding nature of jobs like farming will encourage rural residents to engage in more physical activity [41]. Consequently, this might help prevent weight gain and lower the prevalence of metabolic disorders among pregnant women living in rural areas. Moreover, studies from Bangladesh [43] and China [42] indicate that urban women are more likely than rural women to be overweight or obese. As a result, urban families are more vulnerable to metabolic diseases due to an increased incidence of obesity [42]. Nevertheless, while urban children generally have a lower likelihood of experiencing undernutrition in comparison to their rural counterparts, this circumstance does not apply equally to all urban children, especially those who struggle with poverty. Challenges such as substandard housing, restricted access to services, susceptibility to environmental hazards, and inadequate and inconsistent incomes in densely populated low-income nations exacerbate the issue of food insecurity and malnutrition [44].

Strengths and limitation of the study

This research has several noteworthy strengths. The analysis is based on survey datasets that are nationally representative and have substantial sample sizes. This guarantees that the findings are statistically sound, accurate, and reliable. Furthermore, the DHS included all administrative divisions for the relevant nation, therefore ensuring the generalizability of these results to the whole country. Nevertheless, there are limitations.

Given its cross-sectional design, the survey is unable to evaluate long-term changes in nutritional status. Moreover, relying only on BMI as a parameter for evaluating the nutritional condition of mothers is less reliable in comparison to more precise indicators such as waist-hip ratio and skinfold thickness.

Conclusion

In conclusion, the prevalence of DBM is increasing at the household level in South and South-east Asian countries, especially in urban areas and wealthy households. Although progress has been made in decreasing the prevalence of underweight, the region still faces a substantial problem of undernutrition. Simultaneously, the rising prevalence of obesity presents a raising challenge. Further research is necessary to provide pragmatic remedies and preventive measures to address this complex problem. Future research should give priority to identifying country-specific risk factors that lead to the high prevalence of DBM and the factors that contribute to a higher prevalence of maternal overweight. Furthermore, it is essential to explore risk factors specific to urban and rural locations for DBM, since this study revealed differences between these settings.

Contributors

AT and HS conceptualized and designed the study. AT led the model development, natural history calibration, and was responsible for the collation and integration of data into the model, performed model analysis. AT drafted the manuscript. HS, MK and DG supervised the analysis output. All authors reviewed and edited subsequent drafts. All authors read and approved the final report and made the decision to submit for publication.

Data sharing statement

Data is freely available on the following website: <https://dhsprogram.com/data/>

Declaration of interests

We declare no competing interests.

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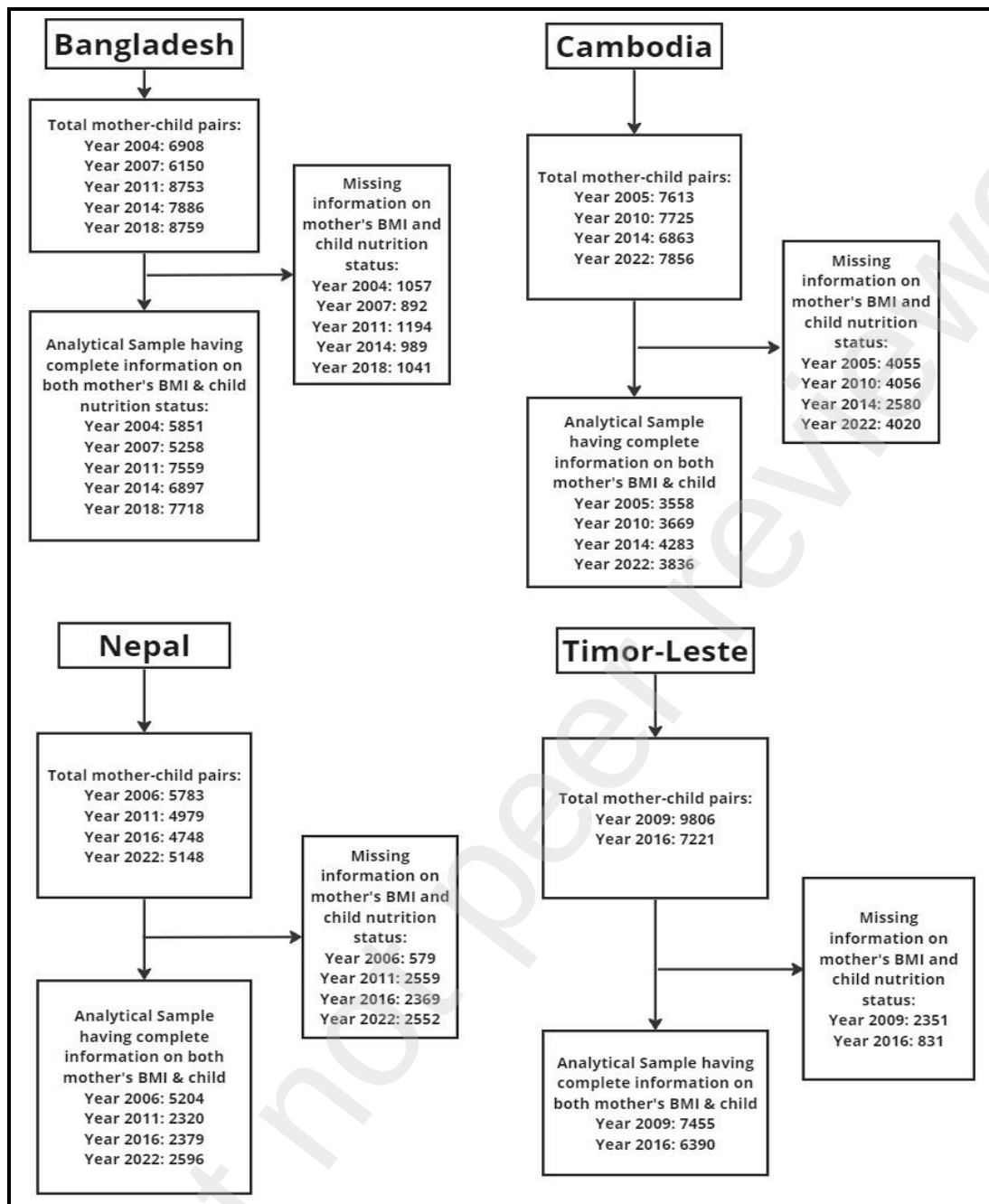


Figure 1: Flow chart showing sample selection procedure

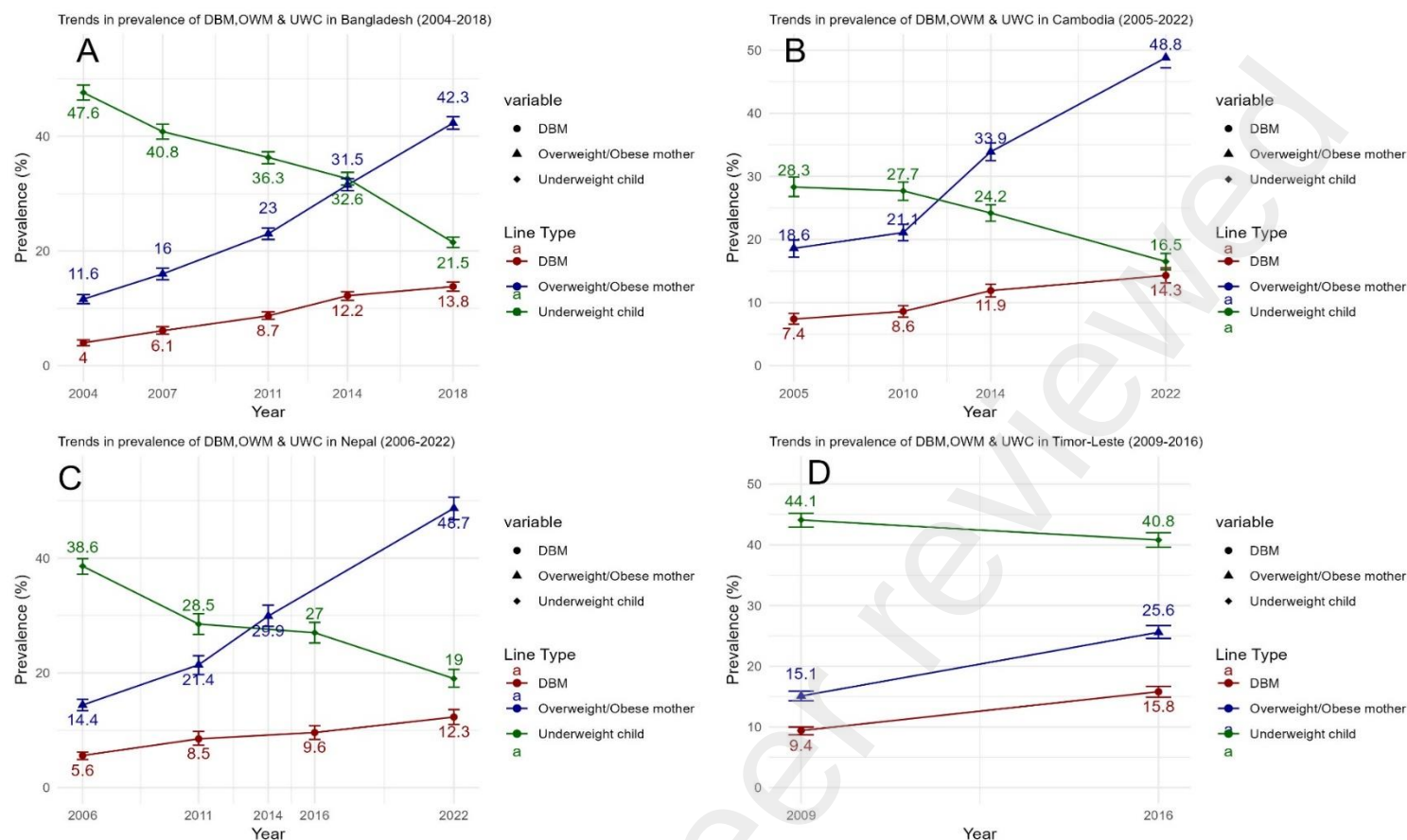


Figure 2: Change in the prevalence of household level DBM among mother–child pairs in Bangladesh (A), Cambodia (B), Nepal (C) & Timor-Leste (D).

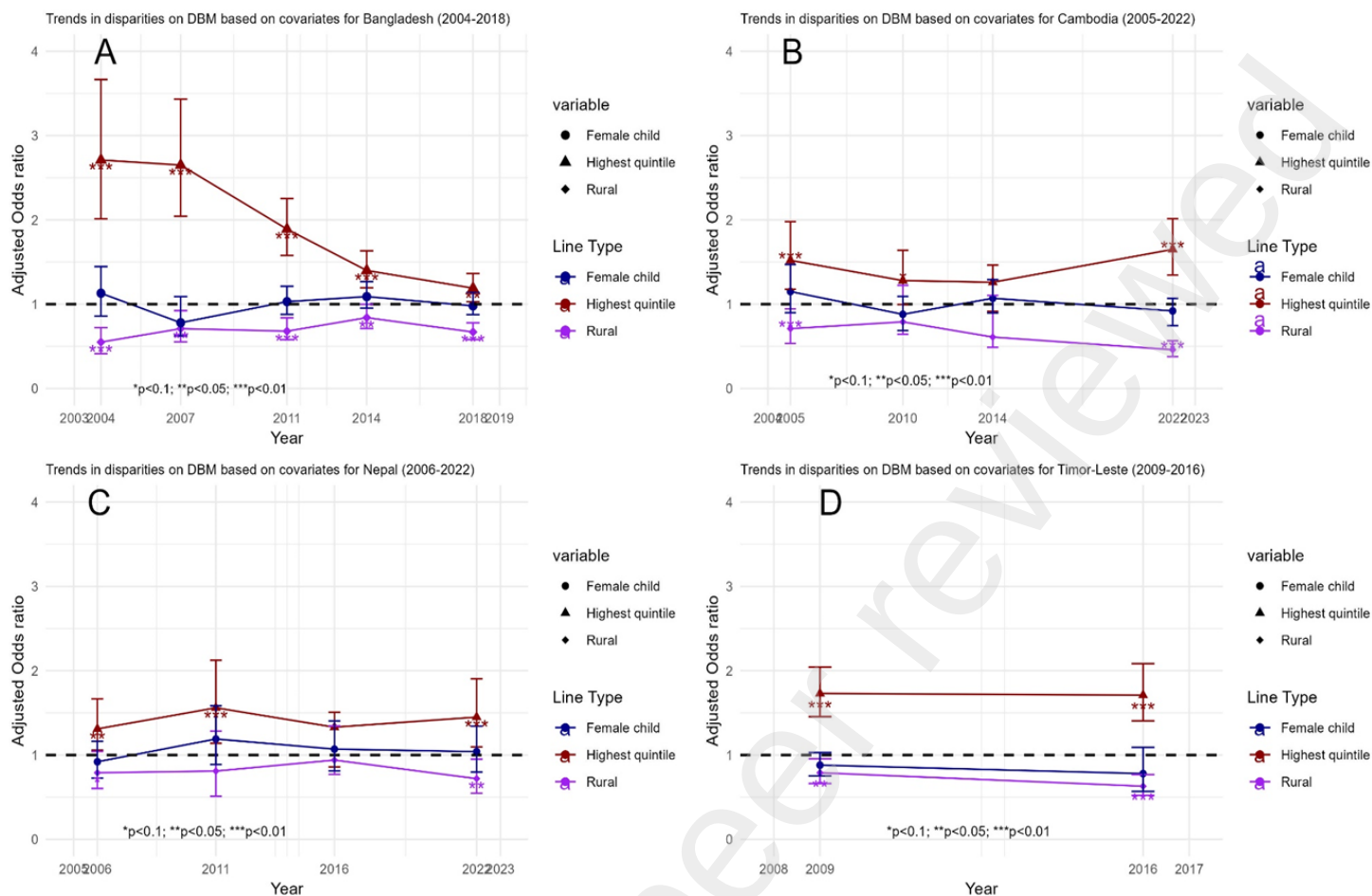


Figure 3: Trends in disparities for household level DBM among mother-child pairs in Bangladesh (A), Cambodia (B), Nepal (C) & Timor-Leste (D) based on wealth index, place of residence, and child sex. Odds ratio was estimated by mutually adjusting for each other, considering wealth index: lowest quintile (Ref) & highest quintile; place of residence: urban (Ref) & rural, and child sex: male (Ref) & female. “Ref” is the reference category for the odds ratio.

Table 1. Frequency distribution with background characteristics of study participants of Bangladesh, Cambodia, Nepal & Timor-Leste

Covariates	Bangladesh					Cambodia				Nepal				Timor-Leste	
	Year: 2004	Year: 2007	Year: 2011	Year: 2014	Year: 2018	Year: 2005	Year: 2010	Year: 2014	Year: 2022	Year: 2006	Year: 2011	Year: 2016	Year: 2022	Year: 2009	Year: 2016
Place of residence															
Urban	30.1	32.9	30.7	31.4	34.9	20.3	26.1	27.4	35.2	22.5	20.4	43.8	49.5	21.6	29.2
Rural	69.9	67.1	69.3	68.6	65.1	79.7	73.9	72.6	64.8	77.5	79.6	56.2	50.5	78.4	70.8
Wealth Index															
Poorest	22.7	19.9	22.1	21.7	22.4	27.3	25.2	23.9	31.4	28.7	30.4	25.1	33.2	23.4	20.1
Poorer	18.9	20.5	19.5	18.8	20.1	23.4	19.8	18.4	18.3	21.1	20.5	21.6	20.9	20.3	20.8
Middle	18.7	18.8	19.2	19.8	18.2	19.4	17.1	16.1	18.3	17.7	19.0	21.5	19.4	20.8	20.3
Richer	18.2	18.7	19.5	20.3	19.8	15.2	17.4	17.7	18.2	18.0	15.3	20.1	15.2	19.8	22.0
Richest	21.4	22.1	19.8	19.3	19.4	14.7	20.5	24.0	13.7	14.4	14.8	11.8	11.3	15.7	16.8
Child sex															
Male	50.8	50.3	51.2	51.3	52.0	49.7	50.8	50.5	51.3	50.6	51.9	52.3	52.4	49.9	51.7
Female	49.2	49.7	48.8	48.7	48.0	50.3	49.2	49.5	48.7	49.4	48.1	47.7	47.6	50.1	48.3
Total Sample	5851	5258	7559	6897	7718	3558	3669	4283	3836	5204	2320	2379	2596	7455	6309

Table 2. Prevalence of household level DBM by covariates among mother-child pairs of Bangladesh, Cambodia, Nepal and Timor-Leste

Covariates	Bangladesh					Cambodia				Nepal				Timor-Leste	
	Year: 2004	Year: 2007	Year: 2011	Year: 2014	Year: 2018	Year: 2005	Year: 2010	Year: 2014	Year: 2022	Year: 2006	Year: 2011	Year: 2016	Year: 2022	Year: 2009	Year: 2016
Place of residence															
Urban	7.7	8.9	12.4	14.8	15.8	10.7	11.1	13.2	19.4	7.1	9.9	10.0	14.2	12.5	20.0
Rural	3.2	5.2	7.6	11.2	11.9	7.1	8.1	11.6	12.1	5.3	7.8	9.6	11.1	8.5	12.8
Wealth Index															
Poorest	2.1	2.9	4.4	8.3	11.1	6.0	6.7	11.2	11.0	5.5	6.4	10.9	11.2	5.7	9.0
Poorer	2.1	3.6	8.1	12.0	10.9	5.9	7.9	12.5	13.7	3.9	5.9	8.0	10.0	7.9	16.6
Middle	3.8	5.0	7.6	13.3	14.5	7.8	9.7	12.4	15.6	4.8	7.0	8.2	13.1	8.8	16.0
Richer	5.6	9.0	12.0	13.6	16.4	10.9	9.7	10.7	14.2	6.0	10.4	10.7	13.6	10.9	18.5
Richest	8.9	11.5	13.9	14.7	13.9	11.1	10.6	14.2	16.5	9.3	14.6	12.5	19.4	15.6	20.0
Child sex															
Male	3.9	6.9	9.0	12.4	13.5	7.4	9.1	12.0	15.4	5.5	7.6	9.7	12.4	10.0	15.5
Female	5.2	6.0	9.2	12.2	13.1	8.3	8.4	12.1	13.9	5.9	9.0	10.0	12.9	8.8	14.2

Note: Each cell in the table denotes the prevalence of DBM among mother-child pairs across different covariates in Bangladesh, Cambodia, Nepal, and Timor-Leste. In both Bangladesh and Timor-Leste, DBM exhibits statistically significant associations with place of residence and wealth index throughout all observed years ($p < 0.05$). In Cambodia, these associations were significant in the years 2005, 2010, and 2022 ($p < 0.05$). In Nepal, the wealth index demonstrates a significant association with DBM across all years, while place of residence shows significance only in 2006 and 2022 ($p < 0.05$). Also, note that the p values were derived from chi-square tests.